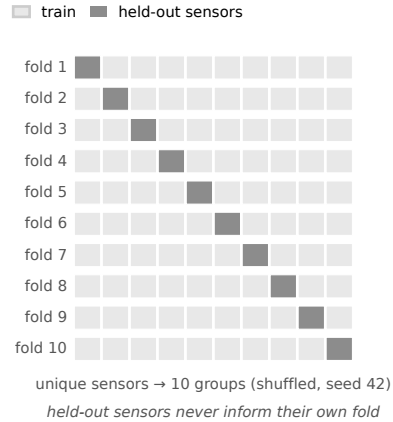
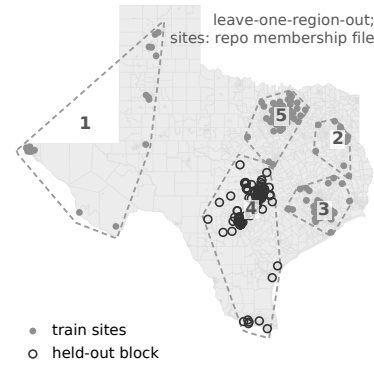


AQNet validation protocol — four axes plus the Tier-3 meta split

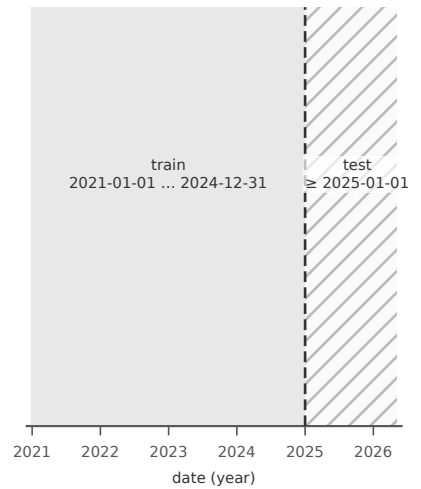
(a) LOSO — GroupKFold(10) over sensors



(b) spatial blocks — KMeans(5)

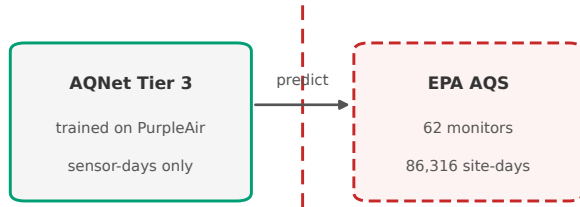


(c) temporal holdout — cutoff 2025-01-01



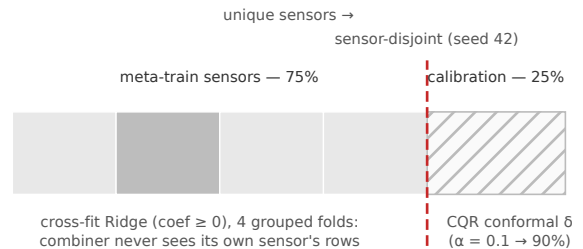
(d) external validation — EPA AQS FRM/FEM

leakage boundary



scored vs FRM/FEM truth — AQS never used in training,
features, stacking, or conformal calibration

(e) Tier-3 meta split — cross-fit vs calibration



one sub-fold at a time is held out inside meta-train (darker block)